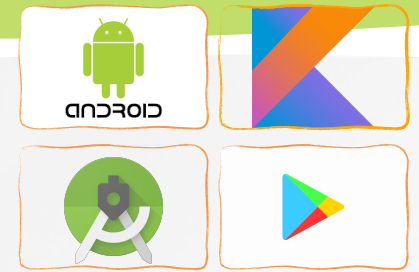




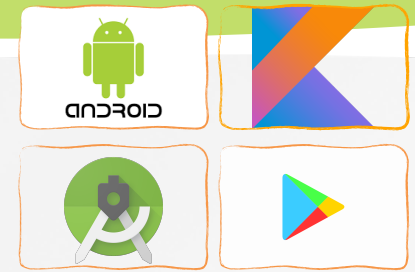
Software Development for Mobile Devices

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Contents

- List-Based Widgets:
 - ListViews and custom ListView
 - RecyclerView
 - Spinner,
 - GridView,
 - AutoComplete TextView
 - HorizontalScrollView
 - **CardView**

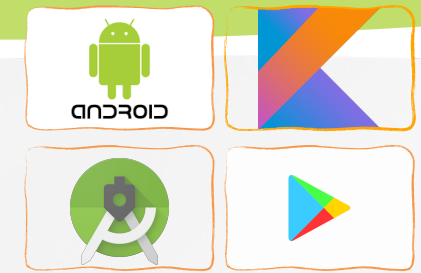


CardView

- Add libs

```
dependencies {  
    //...other libs  
    implementation(libs.androidx.cardview)  
}
```

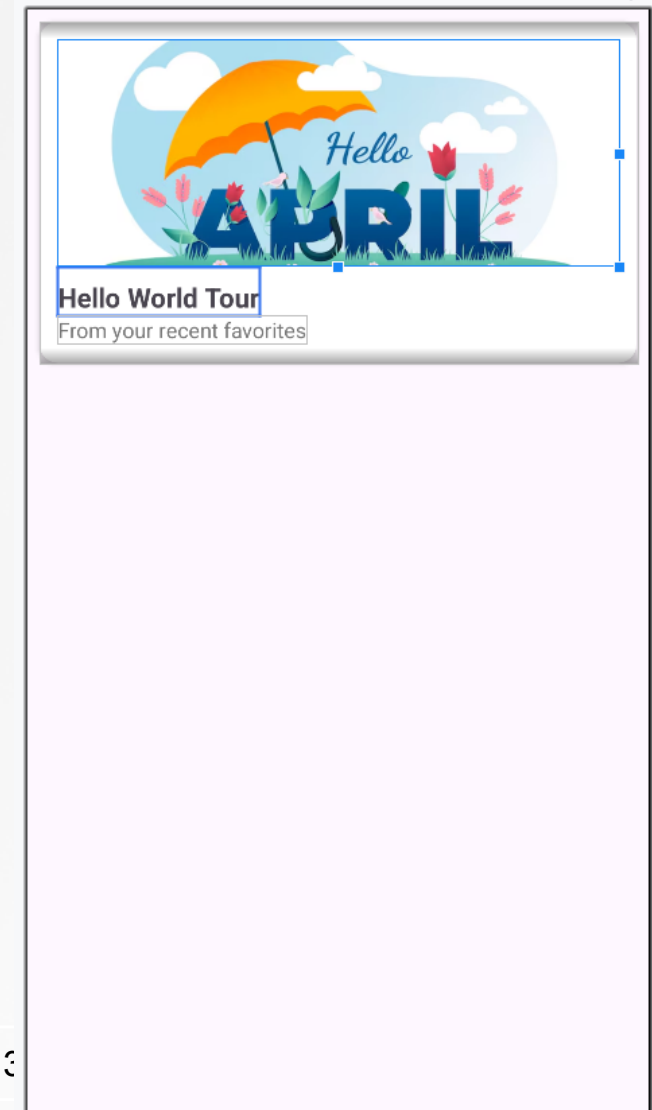
```
<?xml version="1.0" encoding="utf-8"?>  
<androidx.cardview.widget.CardView  
    xmlns:android="http://schemas.android.com/apk/res/android"  
    xmlns:card_view="http://schemas.android.com/tools"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:layout_margin="8dp"  
    card_view:cardCornerRadius="12dp">  
  
</androidx.cardview.widget.CardView>
```

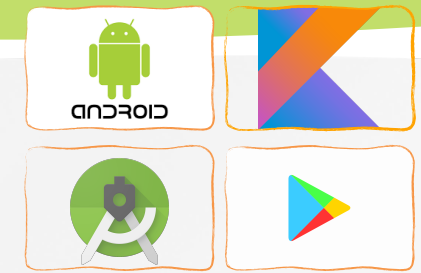


Item card layout

» item_card.xml

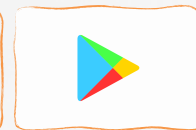
```
... <LinearLayout android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="12dp">
    <ImageView android:id="@+id/image"
        android:layout_width="match_parent"
        android:layout_height="150dp"
        android:scaleType="centerCrop"
        android:src="@drawable/sample01" />
    <TextView android:id="@+id/title"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Glass Souls' World Tour"
        android:textStyle="bold"
        android:textSize="18sp"
        android:paddingTop="8dp" />
    <TextView android:id="@+id/subtitle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="From your recent favorites"
        android:textSize="14sp"
        android:textColor="#777777" />
</LinearLayout>
```





Main layout

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://
schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/recyclerView"
        android:layout_width="0dp"
        android:layout_height="0dp"
        app:layoutManager="androidx.recyclerview.widget.GridLayoutManager"
        app:spanCount="2"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```



Model and Adapter

```
data class CardItem(val imageRes: Int, val title: String, val subtitle: String)

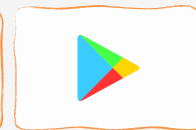
class CardAdapter(private val items: List<CardItem>) :
    RecyclerView.Adapter<CardAdapter.CardViewHolder>() {

    class CardViewHolder(view: View) : RecyclerView.ViewHolder(view) {
        val image: ImageView = view.findViewById(R.id.image)
        val title: TextView = view.findViewById(R.id.title)
        val subtitle: TextView = view.findViewById(R.id.subtitle)
    }

    override fun onCreateViewHolder(parent: ViewGroup, viewType: Int): CardViewHolder
    {
        val view = LayoutInflater.from(parent.context)
            .inflate(R.layout.item_card, parent, false)
        return CardViewHolder(view)
    }

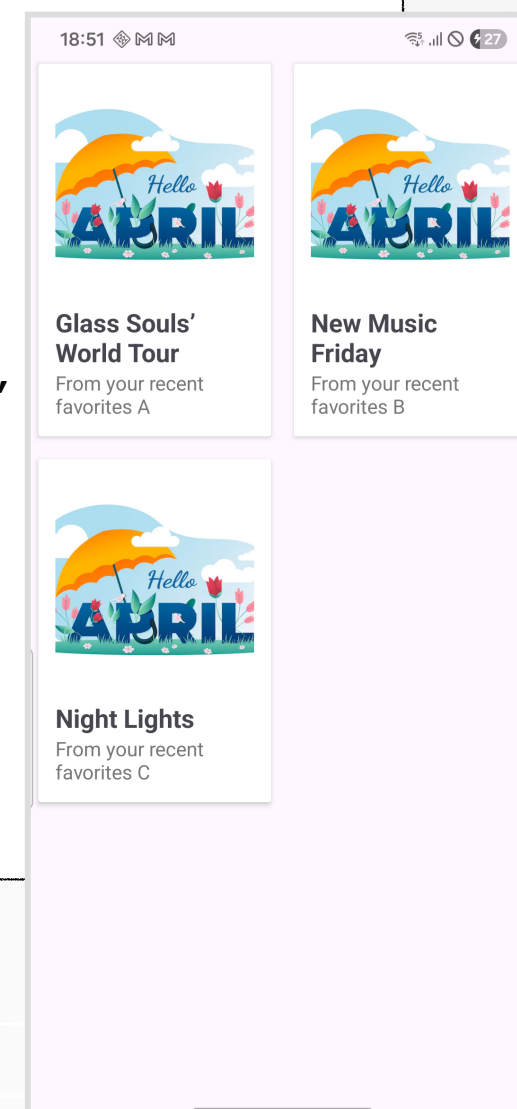
    override fun onBindViewHolder(holder: CardViewHolder, position: Int) {
        val item = items[position]
        holder.title.text = item.title
        holder.subtitle.text = item.subtitle
        holder.image.setImageResource(item.imageRes)
    }

    override fun getItemCount() = items.size
}
```

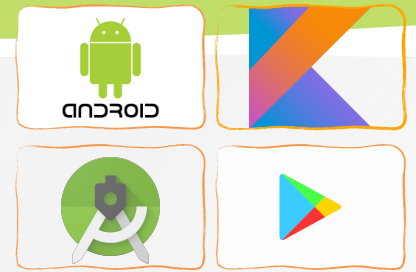


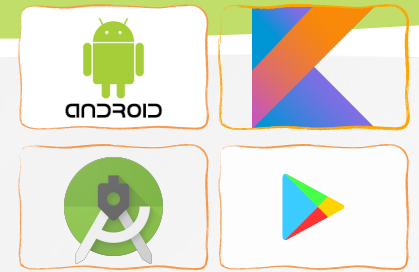
Main activity

```
class MainActivity : AppCompatActivity() {  
    override fun onCreate(savedInstanceState: Bundle?) {  
        super.onCreate(savedInstanceState)  
        //...  
        var recyclerView : RecyclerView  
        recyclerView = findViewById(R.id.recyclerView)  
        recyclerView.layoutManager = GridLayoutManager(this, 2)  
  
        val items = listOf(  
            CardItem(R.drawable.sample01, "Glass Souls' World Tour",  
                "From your recent favorites A"),  
            CardItem(R.drawable.sample01, "New Music Friday",  
                "From your recent favorites B"),  
            CardItem(R.drawable.sample01, "Night Lights",  
                "From your recent favorites C")  
        )  
  
        recyclerView.adapter = CardAdapter(items)  
    }  
}
```



Q&A





Reference

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- » Ted Hagos, Learn Android Studio 3 with Kotlin, 2018
- » Peter Späth, Pro Android with Kotlin, 2019
- » Milos Vasic, Mastering Android Development with Kotlin, 2017
- » Victor Matos, Mobile Application Development course, Cleveland State University
- » <https://developer.android.com/guide/>